

Heavy Duty - 9919

9919 Shepard Road SE
Calgary, Alberta
T2C 4M5
(403) 523 - 0488



Estimate: EST-S2-16541

Estimate date: **Jul 28, 2026**

ADDRESSES

BILL TO

American Heavy Duty Truck Repair
40807 Clark Locks Apt. 699
Apt. 011
South Karen, Nunavut E5T7S1

REMIT PAYMENT TO

**Foothills Group Inc c/o Interstate
Billing Service**
C/O T04220C PO Box 4220, STN A
Toronto, Ontario M5W 3B9
(403) 523 - 0488

ASSET	UNIT	PLATE	MILEAGE	ENG HRS	VIN / SERIAL
2006 Peterbilt 378	UNIT 23	MHL-6397	415,358	17,140	01G7WPFKU CHJJ9RJS

Customer PO **23**

Terms **Net 30**

WORK SUMMARY

01 Service - Perform LOF and inspection

SCOPE OF WORK

TM: Drained old oil and removed old oil filter, reinstalled drain cap and filled new oil filter with new oil and installed. Filled engine with new oil and checked level. Greased all grease fittings. Job completed.

LABOR

Service - Perform LOF and inspection	\$224.92
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PARTS

15W40 Bulk- Mobil 1 Delvac HDEO	42	\$7.13	\$299.46
Windshield Washer Fluid 4L	1	\$7.09	\$7.09
Brake & Parts Cleaner	1	\$7.99	\$7.99
EP2 Grease Tube, 390G	1	\$11.50	\$11.50
NanoNet Engine Oil Filter, X12/X15	1	\$107.02	\$107.02

Line total **\$657.98**

02 Service - Air filter

SCOPE OF WORK

CK: Engine air filter replaced. Job complete.

LABOR

Service - Air filter			\$74.98
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PARTS

Engine Air Filter	1	\$155.16	\$155.16
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Line total **\$230.14**

03 Service - Cabin air filter

SCOPE OF WORK

CK: Replaced cabin air filter. Job complete.

LABOR

Service - Cabin air filter			\$44.98
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PARTS

FILTER ASSY.	1	\$45.03	\$45.03
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Line total **\$90.01**

04 Service - Replace fuel filters

SCOPE OF WORK

TM: Removed old fuel filters, filled new ones with filtered diesel and installed. Job Completed.

LABOR

Service - Replace fuel filters			\$119.96
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PARTS

Fuel Filter, Cummins ISM/ISX	1	\$41.18	\$41.18
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Fuel Filter, Cummins ISX/X15	1	\$37.07	\$37.07
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Line total **\$198.21**

05 Diagnose - Starter

Loud ticking noise prior to fire starting, almost as if the start kept going.
Had a fire beside the starter, customer thinks there is a short.
RO 10062 - we replaced the starter last year

SCOPE OF WORK

JL: Inspected starter. Noted signs of fire at upper starter solenoid. Inspected and noted braided fuel lines are rubbing on solenoid. Suspect steel braided fuel lines have rubbed through to chords, causing starter to short to ground through fuel lines. Recommend removing starter and performing further inspection, advise on fuel line condition. Diagnosis in progress.

CM: Disconnect batteries. Remove all starter cables and mounting bolts. Remove the starter. Cut all zip ties securing hoses and remove loom. Clean off grease and dirt. Inspect the steel braided fuel line coming from the pump and found exposed cords in the same area as the location of the fire at the starter. Further back in the frame rail found the hose leaking fuel. Cut all zip ties securing the fuel line bundle along the frame rail. Disconnect at the rear Tee fitting and remove hose. Inspect the other fuel lines and cables near the starter. Found a steel braided line cut and hanging by the starter. Follow the hose back and found it goes to the air dryer and is not in use. Cut all ties and remove the hose from the truck. No other issues found with the starter cables or fuel lines. The starter and the 3/4" steel braided fuel line from the pump to the rear tee need to be replaced. Diagnosis complete.

LABOR

Diagnose - Starter \$641.79

Line total \$641.79

06 Replace - Batteries Failed load test

SCOPE OF WORK

TM: Replaced batteries with 4 new ones, left 2 negative terminals disconnected for trouble shooting with the starter issue. Job complete.

LABOR

Replace - Batteries \$224.92

PARTS

Group 31 Battery- Stud Post 950CCA	4	\$151.00	\$604.00
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Line total \$828.92

07

Diagnose - Issue starting

Starter and ignition signal is not functioning properly. Will try and start itself without being told to.

SCOPE OF WORK

CM: Begin diagnostics for unit trying to engage starter without turning the key. Reconnect negative battery cables. Hook up the battery charger. Check voltage reading at the S terminal on the starter. Reading 0 volts. Turn key to start, and recheck. 3.5 volts present at S terminal and solenoid is clicking. Indicating low battery voltage present. Turn key off and solenoid continues to activate. Remove negative cable from the batteries. Now when reconnecting the negative cable, power is being sent to the solenoid. Visually inspect cables from the starter to the batteries. No issue found. Suspect failed/shorted batteries causing the erratic starting issues. Remove the positive and negative battery cables. Allow each battery to charge before load testing. A 475 amp load is required to properly load test each battery, and the voltage reading must stay above 10 volts for the battery to pass. After applying a load of 100 amps the voltage reading on the carbon pile drops to 1 volt on all 4 batteries. The batteries need to be replaced. Will need to recheck the operation of the starter and electrical system after battery replacement.

After battery replacement attempt to start. Engine now starts. Shut down engine and restart multiple times with no issue. After continuing to test the starter, found that only turning the key to the ACC position will start the engine, and turning the key off will trigger the solenoid to activate the starter. Disconnect the S-terminal and check for power. 12 volts is now being supplied to the S terminal at all times. Remove the dash trim and ignition. Bench test the ignition. Hook up a power and ground. With key off ACC, IGN, and ST reading 0V. Turn key to ACC. Now the ACC pin is reading 12 volts, no voltage on ignition or ST pin. Turn the key to the run position. Now IGN pin and ACC are reading 12 volts and ST still reading 0V. Turn key to start, ACC reading changes to under 1 volt and ST pin and IGN are supplied with 12 volts. Ignition is ok. Locate the Starter Relay on the fire wall and remove for testing. Bench test relay. Could not find a fault. Install a new relay and retest unit. Truck starts and shuts down normally. 12 volts is no longer being supplied to the S terminal in all positions. Test multiple times. Let unit sit with the key on for monitoring. Cannot duplicate starting faults.

CM: Recheck starter operation. No further issues found. Battery voltage being supplied to the S-terminal at the correct key positions. Secure starter cables, check over job. Found the S-terminal connection slightly loose. Screw will not tighten down. When removing the screw for inspection the threads came out of the solenoid. Remove the solenoid from the unit to repair. Rivet an eye lit connector onto the S- terminal. Connect a new cable, wrap in loom and shrink tube connections. Re install the starter solenoid. Tighten all wire connections and mounting bolts. Resecure wiring. Start unit. Unit will not build air. Check for leaks. Found the pressure relief valve at the air dryer removed for towing. Clean up threads and reinstall. Run unit. Confirm air pressure builds to governor cut out. Park unit outside. Transmission grinds trying to get unit into gear. Created line to diagnose clutch brake not working. Job complete.

LABOR

Diagnose - Issue starting	\$1,106.63
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PARTS

PRIMARY WIRE- 10GA, 12GA,14GA, 16GA, 18GA, 20GA, 22GA (PER FOOT)	1	\$0.68	\$0.68
3/8 BLACK SHRINK TUBE	1	\$2.67	\$2.67

12-10GA SEEMLESS BUTT CONNECTR	2	\$0.85	\$1.70
5/8" SPLIT LOOM	2	\$1.02	\$2.04
RELAY 12V 40/30 AMP - WEATHER	1	\$12.22	\$12.22

Line total **\$1,125.94**

08 Service - Install driveshaft Unit was towed in

SCOPE OF WORK

CM: Unit Towed in with Drive shaft removed at the front differential. Support drive line on a stand. Clean up the U joint and yoke. Inspect U joint and found both trunnions have brinelling marks. The U joint will need to be replaced. Job complete.

LABOR

Service - Install driveshaft	\$112.46
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Line total **\$112.46**

09 Replace - U joint Front Diff input U joint- trunnions have brinelling marks

SCOPE OF WORK

CM: Remove U joint caps, use the tiger tool to remove the U joint from the drive shaft. Clean up drive shaft and yoke. Install a new U joint. Torque cap mounting bolts and strap bolts to spec. Grease U joint. Job complete.

LABOR

Replace - U joint	\$187.44
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PARTS

1810 HALF ROUND U-JOINT	1	\$191.59	\$191.59
STRAP KIT	1	\$29.65	\$29.65

Line total **\$408.68**

10 Replace - Starter Caught fire

SCOPE OF WORK

CK: New starter installed. Three mounting bolts tightened with 1/2 impact, all positive and negative connections done up and secured repspectivly. Job complete.

LABOR

Replace - Starter \$337.39

PARTS

Starter, 11 Tooth, Cummins/Caterpillar	1	\$834.75	\$834.75
Heat Shrinkable Butt Connector, 12-10 Gauge	1	\$1.80	\$1.80

Line total \$1,173.94

11 Replace - Fuel Line Cords exposed and leaking - steel braided line from pump to rear tee

SCOPE OF WORK

CK: Built a new fuel line and ran it from the side of the engine along the frame rail in a manor that avoid the starter and secured the fuel line all the way to the T fitting between fuel tanks. Job complete.

LABOR

Replace - Fuel Line \$382.37

PARTS

#08 SAE100R5 1/2" HYDRAULIC HOSE 2000 PSI	12	\$8.10	\$97.20
3/4" SPLIT LOOM	25	\$1.25	\$31.25
#08 R5 FIELD ATTACHABLE X #08 FEMALE JIC SWIVEL	2	\$16.88	\$33.76

Line total \$544.58

12 Sublet - Tow Tow unit

SCOPE OF WORK

Sublet

LABOR

Sublet - Tow \$0.00

PARTS

Yard Move	1	\$115.39	\$115.39
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Line total \$115.39

13 Quality control check over QC

SCOPE OF WORK

JC: Performed quality control inspection. Job complete.

LABOR

Quality control check over

Line total \$0.00

SUMMARY

Any warranties on the parts and accessories sold hereby are made by the manufacturer. You understand and agree that we make no warranties of any kind unless expressed in writing. You hereby authorize us to perform the repair work herein set forth and to purchase the necessary material and parts to perform such repair work. You agree that we are not responsible for loss or damage to your vehicle or articles left in your vehicle in case of fire, theft, or any other cause beyond our control or for any delays caused by unavailability of parts or delays in part shipments by the supplier or transporter. In addition, you agree that we are not responsible for damages to your vehicle from freezing due to lack of antifreeze. You hereby grant our employees permission to operate your vehicle on streets, highways, or elsewhere for the purpose of testing and/or inspection. You acknowledge and agree that an express mechanic's lien on your vehicle is granted to secure payment of this invoice for the repair work detailed in this invoice.

Labor	\$3,457.84
Parts	\$2,670.20
Shop supplies (10.5% of labor)	\$350.00
Subtotal	\$6,478.04
GST (5%)	\$323.90
Total	\$6,801.94

ESTIMATED TOTAL \$6,801.94

CUSTOMER SIGNATURE

PRINTED NAME

DATE